INSIGHTWELLNESS-AI

Obesity Level Prediction using Machine Learning

What is InsightWellness AI ?

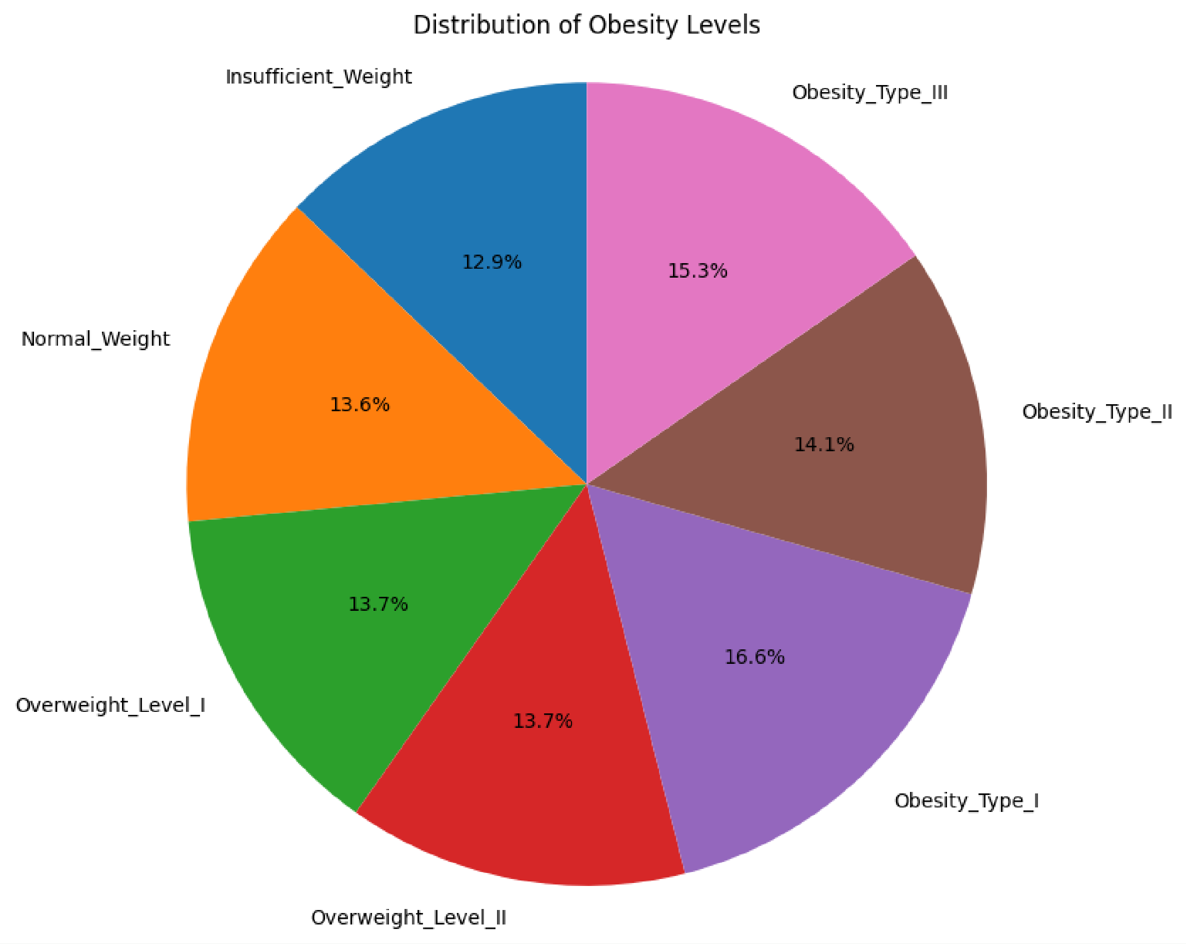
Nowadays, obesity early is essential as it has become one of the most global health challenge. However, many cases are undetected until complications appear later. To address this issue, we propose InsightWellness AI, a machine learning based application designed to predict obesity risk from health and lifestyle data, enabling early intervention.

Dataset overview

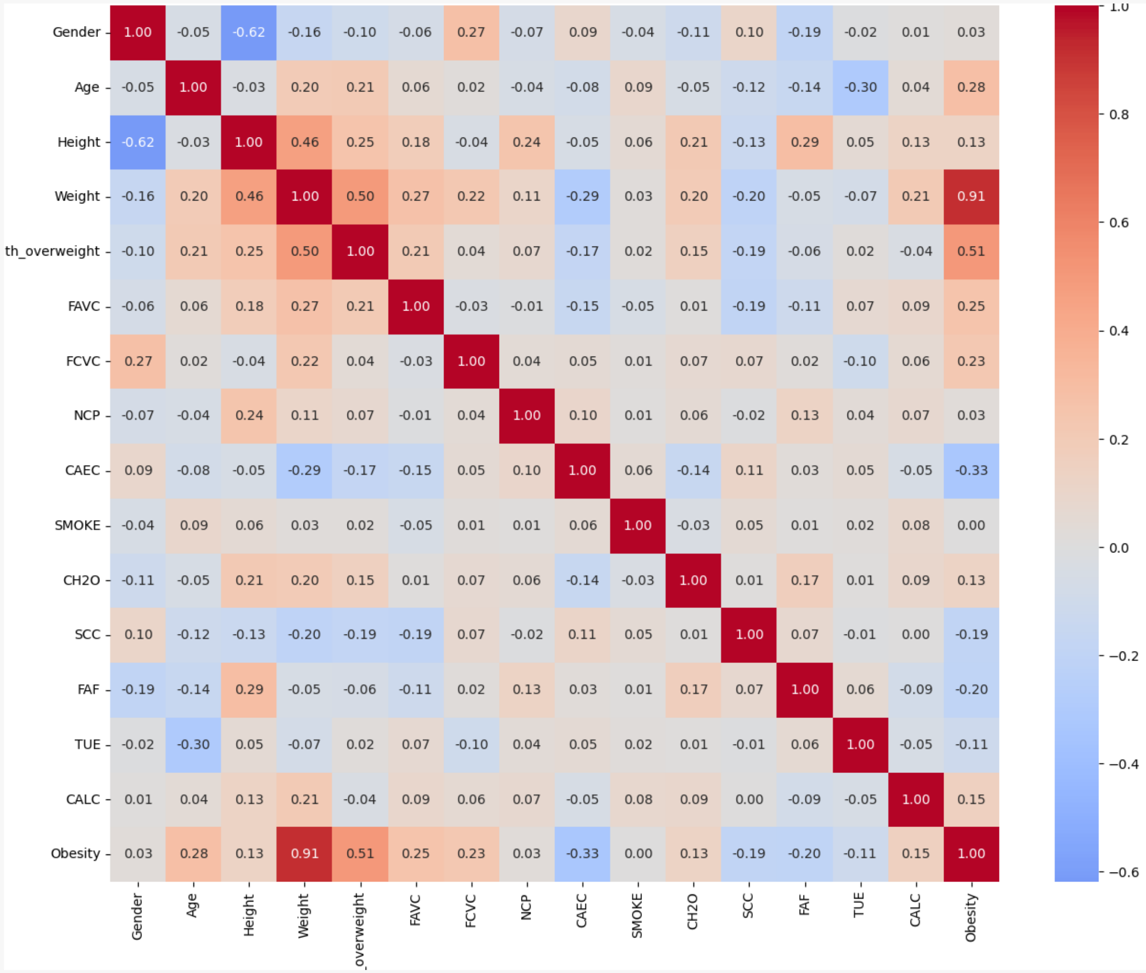
- Source: UCI Machine Learning Repository
- This dataset was created to support the automated estimation of obesity levels based on individuals eating habits and physical condition. Data was collected from individuals in Mexico, Peru, and Colombia via a web survey. To expand the dataset, 77% of the records were synthetically generated using the SMOTE (Synthetic Minority Oversampling Technique) algorithm. Only 23% are original survey responses.
- 2,111 records: 16 features and 1 target
- Target: 7 obesity categories from insufficient weight to obesity type 3.

Exploratory data analysis.

- No missing values.
- 24 duplicate rows were detected (1.14%).
- The dataset is approximately balanced across the 7 classes:



Correlation matrix



- Weight (0.91) is by far the most correlated feature with obesity levels.
- Family history with overweight (0.51) has a moderate correlation.
- Age (0.28), high-calorie food consumption (0.25) and vegetable consumption (0.23) are positively correlated.
- Physical activity frequency (-0.20), calorie monitoring (-0.19) and eating between meals (-0.33) are negatively correlated.
- Other variables show little correlation.

Data preprocessing

- Ordinal encoding: CAEC and CALC (No=0 to Always=3)
- Binary encoding for gender, FAVC, smoke, SCC and family history with overweight.
- Target encoding: the 7 classes are mapped to 0-6.
- Target: 7 obesity categories from insufficient weight to obesity type 3.
- One-hot encoding for MTRANS (reference class dropped).
- Storage: Raw, preprocessed and split data are stored on Google cloud storage.

Model Comparison

Model	Accuracy	F1-macro	CV F1 mean	CV F1 std
Decision Tree	92.2%	91.91%	91.12%	±0.85%
Random Forest	94.56%	94.49%	94.33%	**±0.43%**
Gradient Boosting	**96.45%**	**96.33%**	**96.18%**	±0.93%

- 3 models evaluated
  - Decision tree
  - Random Forest
  - Gradient Boosing
- 5-fold cross-validation to assess statibility

Gradient boosting and Hyperparameter tuning

Gradient boosting is better than both alternatives across the different metrics and several random states.

For the tuning a grid search with 5-fold CV on training set has been used with:
- 500 estimators
- Max depth $\in [3, 5, 7]$
- Learning rate $\in [0.05, 0.1, 0.2]$

Parameter	Value
max_depth	5
Learning rate	0.05

Final Model Performance

Final Model Results

Metric	Value
Test Accuracy	95.51%
F1-macro	95.38%
Precision	95.70%
Recall	95.33%
CV F1 mean	96.73%
CV F1 std	±0.7%

Top 5 feature importances

Feature	Importance
Weight	51.94%
Height	17.04%
FCVC	9.09%
Gender	7.20%
Age	3.81%

Training emissions: 0.000158 kg CO_2eq tracked with CodeCarbon

MLOPs setup

- Version control: GitFlow
- CI/CD: pre-commit hook, ruff and pytest placeholder (lab1).
- Data storage: Google cloud storage.
- Everything done with data on GCS, no local files.

Next steps

- Build a REST API to serve the model
- Package the API in a Docker container
- Deploy to Google Cloud